



Madduri Lakshmi Naga Sai Srivalli

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PROFILE

I'm a Passionate Data Science Engineer exclusively into building end to end models for various use cases involving Computer Vision and Natural Language Processing. Have expertise in crafting precise and effective AI prompts to optimize language model outputs and drive innovative solutions. I'm a vision driven individual and I strive to reach to roots in research, quick and efficient in implementations. I proved these skills in various projects at numerous circumstances across my tenure.

EMPLOYMENT HISTORY

Jul 2023 — Present

Graduate Engineer Trainee, Orange Business

Gurgaon

InfoGenie: QA Bot

Project Overview: Developed and implemented an intelligent Question-Answering bot designed to resolve questions related to various services within the company for easy cross team communication and reduced on boarding effort into new teams.

Implementation: Implemented this QA Bot using a combination of prompts with different versions of gpts (gpt4, gpt4o). The key idea is to first identify which team and service the question is about and then retrieve and send the corresponding information using RAG to our main Answering prompt which uses the retrieved information to generate the most suitable answer in either English or French. There are dedicated prompts to identify the Team, identify the Service and identify language and translation prompt along with the RAG and main answer generation prompt. All these blocks when combined have produced reliable results in answering basic to intermediate level questions accurately.

Gerez-Vous: Meeting Summarizes

Project Overview: It is a Manager assisting bot built using Large Language Models that helps to prepare minutes of the meeting using the transcript of a team's meeting. A Flask application for the same has been made and hosted on a flask server.

Model Selection: Various Decoder based Models starting from TF-IDF to transformer based models like bart, T5, gpt2-medium, gpt2-large, gpt3.5, gpt4 were used before choosing the best performing model in the expected budget. Till gpt-3.5, due to context window limitations, input text was split into chunks and processed sequentially. To maintain context, a portion of the previous output was included with each subsequent chunk before summarization, with results then concatenated for the final output. However, the limited context window didn't help the self-attention mechanism of the LLMs to grab the whole context of the team's meeting. gpt-4 is the best performing model among the researched models for summarisation.

SmartAgri : Agricultural Pest Recognizer

Project Overview: Developed a Deep Learning recognition model to help farmers identify pests in agricultural fields with a confidence score indicating the probability of the detected pest belonging to a particular class.

Dataset: The Dataset Comprised of 1518 pest images uniformly distributed over 5 classes. 1100 images used for training and 418 images used for testing.

Model Selection and Evaluation: Chose YOLO-V3 a variation of Dark-net with 106 layers architecture, pre-trained on the COCO dataset. YOLO-V3 is chosen for its dual capability of performing both detection and recognition tasks, fast object detection algorithm and it's light weight nature to deploy in a device like JETSON Nano suitable for field use due to its limited capabilities.

Metrics:

- Recall ranged from 88% to 100% (with a "None" class for unclassified pest images ensuring a 100% recall).
- mean AP achieved was 24.9% and confidence score for the input image belonging to a particular class during inference ranged between 0.78 to 0.99.

Brain segmentation

Project Overview: Developed a Hybrid U-net Transformer (HUT) architecture for lesion segmentation on brain MRI and CT scan data.

Model Selection: UNet alone uses layers of CNN's at its encoders and decoders. CNN's are good at finding pixel relationships at local level but there's a problem of bottle neck i.e. after certain layers of convolution, if the image is convoluted further, results in a huge information loss. This is where visual transformers come into the play, in general, transformers are great at finding global relationships of pixels. Whereas transformers need a very huge amount of data to make them understand local relationships. Therefore, this HUT network uses the advantages of both CNN's and visual transformers i.e CNN's to find the local relationships, and visual transformers to understand the global relationships of the encoded pixels.

Novelty: Added a new module to extract boundary information using transformation of image into frequency

domain to obtain high frequency changes in the image and use them to enrich separation in segmentation of lesions.

Metrics: We reproduced an industry standard research paper and surpassed their performance by 16bps in DSC (Dice Coefficient) and 24 bps in CE(Coverage Efficiency).

Aug 2022 — May 2023

Software Intern, Tidalwave Technologies

Mumbai(Remote)

- Created an inhouse version of string.h library libsst, to avoid importing the heavy module and to improve performance.
- Understood 3GPP International standards of working and implementation of 5G/4G LTE.
- Converted 3GPP's .asn file to .c using the asn1c compiler generating 100s of .c files.
- Wrote a Library libRRC.a that encodes or decodes RRC Messages (RRCConnectionSetup, MasterInformationBlock, RRCConnectionEstablishment, etc) that supports base stations and handheld devices. An external program was able to use this library to encode/decode RRC messages using make or cmake build system in linux based environment.
- The messages themselves belonged to specific RRC channels (eg. BCCH, CCCH, PCCH etc). Messages belonging to a specific channel were contained inside the channel's container/PDU and then encoded/decoded as per the channel's type descriptor.

EDUCATION

Jul 2019 — Jul 2023

B.Tech in Computer Science, Shri Mata Vaishno Devi University

CGPA: 8.43/10

Jammu

RecognizeMe: Multi-View Image Classification of Humans March 2022 – June 2022

- Built a sophisticated multi-input CNN model that recognizes a person when an image with combinational (right & front or left & front) profiles of his face is the input. The model is trained with three profiles (Left, right, front) and tested with an image consisting any combination of the trained profiles.
- Achieved 86% Accuracy on manually collected data of size 900. A data of 20 individuals is collected which includes 13 sets of an individual with three of their facial profiles. 3 parallel pre-trained resNet34 Networks and a classification deep neural network is used while building the model.
- This idea is inspired from a real life scenario where humans can recognize a known person, looking at just 1-2 profiles.

Grow More: A Machine Learning Model for suitable crop prediction.

- Built a Crop Prediction ML model that predicts the best suitable crop for farmers' fields based on soil nutrient concentration values and various weather factors.
- The project is in line with the Indian government's efforts to bring technology in the aid of farmers and specially intended for the farmers with the soil health cards.
- Achieved an accuracy of 98.67% using Random Forest Algorithm on our existing dataset. Successfully built a robust model which performed with significant accuracy even with a lower train to test ratio i.e. 2:8. The performance of our final Random Forest model is compared with other popular models such as Ensemble models, K- Neighbors, PCA SVM to find the best suitable model.

SKILLS

Machine Learning Libraries: Openai, Langchain, LangGraphs, Keras, Tensorflow, PyTorch, Scikit-learn, Numpy, Matplotlib, Pandas, Nltk, Transformers

Softwares and Platforms: MATLAB, Conda, Jupyter Notebook, VS Code, Linux, Data Bricks, GCC compiler, Make/CMake, Flask, Git

Programming Languages: Python, C, Java

Additional Skills: Prompt Engineering, Data Structures and Algorithms

Soft Skills: Cross team projects have evolved new skills in me like project planning and execution, Data Analysis and Visualization, Communication, Presentation and Documentation.

EXTRA-CURRICULAR ACTIVITIES

Carnatic Classical Music & Kuchipudi Classical Dance 15 years of training

- Enrolled in **Guinness Book of World Records twice (2010, 2012)**, for the participation in Maha Brinda Natyam held at Hyderabad, Conducted by Silicon Andhra, USA.
- CCRT Scholarship holder from MHRD- GOI since 2013 in Kuchipudi.
- Diploma in Kuchipudi dance (Distinction division) from Potti Sriramulu Telugu University.
- Diploma in Carnatic Music (First class division) from Potti Sriramulu Telugu University.